Exercise of Supervised Learning: SVM Part 2

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Exercise 1: Kernelized Multiclass SVM

For a data set $\mathcal{D} = \{(\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)})\}$ with $y^{(i)} \in \mathcal{Y} = \{+1, -1\}$, assume that we are provided with a suitable feature map $\phi : \mathcal{X} \to \Phi$, where $\Phi \subset \mathbb{R}^d$. In the featureized SVM learning problem we are facing the following optimization problem:

$$\min_{\boldsymbol{\theta}, \theta_0, \zeta^{(i)}} \frac{1}{2}\boldsymbol{\theta}^\top \boldsymbol{\theta} + C \sum_{i=1}^{n} \zeta^{(i)}$$

$$\text{s.t. } y^{(i)} \left(\left\langle \boldsymbol{\theta}, \phi(\mathbf{x}^{(i)}) \right\rangle + \theta_0 \right) \geq 1 - \zeta^{(i)} \qquad \forall i \in \{1, \dots, n\},$$

$$\text{and } \zeta^{(i)} \geq 0 \qquad i \in \{1, \dots, n\},$$

where $C \geq 0$ is some constant.
(a) Argue that this is equivalent to the following ERM problem:

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{2}||\boldsymbol{\theta}||^2 + C \sum_{i=1}^{n} \max(1 - y^{(i)}(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0).$$

i.e., the regularized ERM problem for the hinge loss for the hypothesis space

$$\mathcal{H} = \{f : \Phi \to \mathbb{R} \mid f(\boldsymbol{z}) = \boldsymbol{\theta}^\top \boldsymbol{z} + \theta_0, \boldsymbol{\theta} \in \mathbb{R}^d, \theta_0 \in \mathbb{R}\}$$

1(a): Rewriting the Optimization Target

Optimization target:

$$\min_{\boldsymbol{\theta}, \theta_0, \zeta^{(i)}} \frac{1}{2}\boldsymbol{\theta}^\top \boldsymbol{\theta} + C \sum_{i=1}^{n} \zeta^{(i)}$$

$$\text{s.t. } \zeta^{(i)} \geq 1 - y^{(i)} \left(\langle \boldsymbol{\theta}, \phi(\mathbf{x}^{(i)}) \rangle + \theta_0 \right), \quad \forall i,$$

$$\text{and } \zeta^{(i)} \geq 0, \quad \forall i.$$

1(a): Comparison between Optimization Target and $\mathcal{R}_{\text{emp}}$

Optimization target:

$$\min_{\boldsymbol{\theta},\theta_0,\zeta^{(i)}} \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{\theta} + C\sum_{i=1}^{n}\zeta^{(i)}$$

$$\text{s.t. } \zeta^{(i)} \geq 1 - y^{(i)}\left(\langle\boldsymbol{\theta}, \phi(\mathbf{x}^{(i)})\rangle + \theta_0\right), \quad \forall i,$$

$$\text{and } \zeta^{(i)} \geq 0, \quad \forall i.$$

Empirical risk:

$$\mathcal{R}_{\text{emp}}(\theta) = \frac{1}{2}\|\theta\|^2 + C\sum_{i=1}^{n}\max(1 - y^{(i)}(\theta^\top\phi(\mathbf{x}^{(i)}) + \theta_0), 0).$$

Observation: Both contain $\frac{1}{2}\theta^\top\theta$. Both contain $C\sum_{i=1}^{n}\ldots$ penalty terms.

Next: Prove that $\zeta^{(i)}$ equals $\max(\ldots)$ term.

1(a): Comparison between Optimization Target and $\mathcal{R}_{\textbf{emp}}$

Optimization target:

$$\min_{\boldsymbol{\theta}, \theta_0, \zeta^{(i)}} \frac{1}{2} \boldsymbol{\theta}^\top \boldsymbol{\theta} + C \sum_{i=1}^{n} \zeta^{(i)}$$

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Empirical risk:

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{2} \|\boldsymbol{\theta}\|^2 + C \sum_{i=1}^{n} \max(1 - y^{(i)}(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0).$$

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1(a): Comparison between Optimization Target and $\mathcal{R}_{\text{emp}}$

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\begin{aligned}
&\min_{\boldsymbol{\theta},\theta_0,\zeta^{(i)}} \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{\theta} + C\sum_{i=1}^{n}\zeta^{(i)} \\
&\text{s.t. } \zeta^{(i)} \geq 1 - y^{(i)}\left(\langle\boldsymbol{\theta}, \phi(\mathbf{x}^{(i)})\rangle + \theta_0\right), \quad \forall i, \\
&\text{and } \zeta^{(i)} \geq 0, \quad \forall i.
\end{aligned}
$$

Empirical risk:

$$
\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{2}\|\boldsymbol{\theta}\|^2 + C\sum_{i=1}^{n}\max(1 - y^{(i)}(\boldsymbol{\theta}^\top\phi(\mathbf{x}^{(i)}) + \theta_0), 0).
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Observation: Both contain $\frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{\theta}$. Both contain $C\sum_{i=1}^{n}\ldots$ penalty terms.

Next: Prove that $\zeta^{(i)}$ equals $\max(\ldots)$ term.

1 (a): Prove $\zeta^{(i)}$ Equals $\max(\ldots)$ Term

For each i, The constraints in the optimization problem:

$$\zeta^{(i)} \geq \underbrace{1 - y^{(i)}\left(\langle \boldsymbol{\theta}, \phi(\mathbf{x}^{(i)})\rangle + \theta_0\right)}_{\text{(i)}}$$

$$\zeta^{(i)} \geq \underbrace{0}_{\text{(ii)}}$$

$\zeta^{(i)} \geq$ (i) and (ii) $\Rightarrow \zeta^{(i)} \geq$ the larger term in (i) and (ii) $\Rightarrow \zeta^{(i)} \geq \max(\text{(i)}, \text{(ii)})$.

Therefore, the constraints translate to $\zeta^{(i)} \geq \max\left(1 - y^{(i)}\left(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0\right), 0\right)$

Note: Our target is to $\min_{\boldsymbol{\theta}, \theta_0, \zeta^{(i)}} \frac{1}{2}\boldsymbol{\theta}^\top \boldsymbol{\theta} + C\sum_{i=1}^{n} \zeta^{(i)}$, so smaller $\zeta^{(i)}$ are preferred. $\rightsquigarrow$ Choose $\zeta^{(i)} = \max(\ldots, 0)$

recall SVM Part 1, Ex. 1(b)

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recall SVM Part 1, Ex. 1(b)

1 (a): Choose $\zeta^{(i)} = \max(\ldots, 0)$

The optimization target

$$\frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{\theta} + C\sum_{i=1}^{n}\zeta^{(i)}$$

becomes

$$\frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{\theta} + C\sum_{i=1}^{n}\max\left(1 - y^{(i)}\left(\boldsymbol{\theta}^\top\phi(\mathbf{x}^{(i)}) + \theta_0\right), 0\right)$$

which is exactly $\mathcal{R}_{\text{emp}}$. $\square$

Exercise 1 (b)

(b) Now assume we deal with a multiclass classification problem with a data set $\mathcal{D} = \{(\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)})\}$ such that $y^{(i)} \in \mathcal{Y} = \{1, \dots, g\}$ for each $i \in \{1, \dots, n\}$. In this case, we can derive a similar regularized ERM problem by using the multiclass hinge loss (see Exercise Sheet 4(b)):

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \frac{1}{2}||\boldsymbol{\theta}||^2 + C \sum_{i=1}^{n} \sum_{y \neq y^{(i)}} \max(1 + \tilde{\boldsymbol{\theta}}^\top \psi(\mathbf{x}^{(i)}, y) - \tilde{\boldsymbol{\theta}}^\top \psi(\mathbf{x}^{(i)}, y^{(i)}), 0),$$

where $\tilde{\boldsymbol{\theta}} := (\theta_0, \boldsymbol{\theta}^\top)^\top \in \mathbb{R}^{d+1}$, and $\psi : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^{d+1}$ is a suitable (multiclass) feature map. Specify a ψ such that this regularized multiclass ERM problem coincides with the regularized binary ERM problem in (a).

1 (b): Think Before Solving

$$\text{(a) binary:} \quad \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \tfrac{1}{2}\|\boldsymbol{\theta}\|^2 + C\sum_{i=1}^{n} \max(1 - y^{(i)}(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0)$$

$$\text{(b) multiclass:} \quad \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \tfrac{1}{2}\|\boldsymbol{\theta}\|^2 + C\sum_{i=1}^{n} \sum_{y \neq y^{(i)}} \max(1 + \tilde{\boldsymbol{\theta}}^\top \psi(\mathbf{x}^{(i)}, y) - \tilde{\boldsymbol{\theta}}^\top \psi(\mathbf{x}^{(i)}, y^{(i)}), 0)$$

- **Multiclass.** SVM is binary by design — one-vs-one, one-vs-rest, or one joint loss?
- **Intercept.** Margin needs $+\theta_0$, but ψ gives only an inner product $\langle \tilde{\theta}, \psi(\mathbf{x}^{(i)}, y) \rangle$. How to absorb θ_0?
- **Equivalence.** Why do (a) and (b) coincide in the binary case ($g = 2$)?

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1(b): A Closer Look at the Multiclass Hinge Loss

Goal: multiclass hinge $\rightarrow$ binary hinge when $g = 2$.

- Binary encoding: $y^{(i)} \in \{1, 2\} \mapsto \{-1, +1\}$.
- Then $\sum_{y \neq y^{(i)}}$ collapses to **one term** ($y = -y^{(i)}$).
- $\psi(\mathbf{x}^{(i)}, y^{(i)})$ uses *both* $\mathbf{x}^{(i)}$ and $y^{(i)}$ — unlike $\phi(\mathbf{x}^{(i)})$.
- No θ_0 in the multiclass loss $\Rightarrow$ absorb it how? *(next)*

Roadmap

Multiclass
one joint hinge loss

Intercept (θ_0)
dummy feature 1

Equivalence
g=2 $\Rightarrow$ binary (a)

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one joint hinge loss

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dummy feature 1

Equivalence
g=2 $\Rightarrow$ binary (a)

1(b): Define $\psi : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^{d+1}$

Goal: write the margin $y(\boldsymbol{\theta}^\top \phi(\mathbf{x}) + \theta_0)$ as one inner product.

1. Absorb θ_0 via a dummy feature 1:

$$\boldsymbol{\theta}^\top \phi(\mathbf{x}) + \theta_0 = \underbrace{\begin{pmatrix} \theta_0 \\ \boldsymbol{\theta} \end{pmatrix}^\top}_{\tilde{\boldsymbol{\theta}}^\top} \underbrace{\begin{pmatrix} 1 \\ \phi(\mathbf{x}) \end{pmatrix}}_{\tilde{\phi}(\mathbf{x})}$$

2. Pull in y: $y\,\tilde{\boldsymbol{\theta}}^\top \tilde{\phi}(\mathbf{x}) = \tilde{\boldsymbol{\theta}}^\top \left(y\,\tilde{\phi}(\mathbf{x})\right)$

3. Scale by $\frac{1}{2}$: $\boxed{\psi(\mathbf{x}, y) = \frac{1}{2}\,y\,\tilde{\phi}(\mathbf{x})}$

Roadmap

✓ **Multiclass**
one joint hinge loss

▶ **Intercept (θ_0)**
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▶ Intercept (θ_0)
dummy feature 1

Equivalence
g=2 $\Rightarrow$ binary (a)

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Roadmap

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one joint hinge loss

▶ **Intercept (θ_0)**
dummy feature 1

Equivalence
$g{=}2 \Rightarrow$ binary (a)

1 (b): What We Prove Next

With $\psi(\mathbf{x}, y) = \frac{1}{2}\, y\, \tilde{\phi}(\mathbf{x})$, it remains to show — in the binary case ($g = 2$) — that the multiclass penalty reduces to the binary hinge of (a):

$$\underbrace{\sum_{y \neq y^{(i)}} \max(1 + \tilde{\boldsymbol{\theta}}^\top \psi(\mathbf{x}^{(i)}, y) - \tilde{\boldsymbol{\theta}}^\top \psi(\mathbf{x}^{(i)}, y^{(i)}), 0)}_{\text{(b) multiclass}}$$

$$= \underbrace{\max(1 - y^{(i)}(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0)}_{\text{(a) binary}}.$$

Roadmap

✓ Multiclass
one joint hinge loss

✓ Intercept (θ_0)
dummy feature 1

▶ Equivalence
g=2 $\Rightarrow$ binary (a)

1 (b): Reduce to the Binary Hinge

Binary case $\Rightarrow$ the sum has **one term** ($y = -y^{(i)}$). Then:

$$
\begin{aligned}
&1 + \tilde{\theta}^\top \psi(\mathbf{x}^{(i)}, y) - \tilde{\theta}^\top \psi(\mathbf{x}^{(i)}, y^{(i)}) \\
&= 1 + \tfrac{1}{2} y\, \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}) - \tfrac{1}{2} y^{(i)}\, \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}) \\
&= 1 + \tfrac{1}{2}(y - y^{(i)})\, \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}) \\
&= \begin{cases} 1 + \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}), & y^{(i)} = -1 \\ 1 - \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}), & y^{(i)} = +1 \end{cases} \\
&= 1 - y^{(i)}\, \tilde{\theta}^\top \tilde{\phi}(\mathbf{x}^{(i)}).
\end{aligned}
$$

Roadmap

- ✓ **Multiclass** — one joint hinge loss
- ✓ **Intercept (θ_0)** — dummy feature 1
- ▶ **Equivalence** — $g{=}2 \Rightarrow$ binary (a)

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\begin{aligned}
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&= 1 + \tfrac{1}{2} y \, \tilde{\boldsymbol{\theta}}^\top \tilde{\phi}(\mathbf{x}^{(i)}) - \tfrac{1}{2} y^{(i)} \, \tilde{\boldsymbol{\theta}}^\top \tilde{\phi}(\mathbf{x}^{(i)}) \\
&= 1 + \tfrac{1}{2} (y - y^{(i)}) \, \tilde{\boldsymbol{\theta}}^\top \tilde{\phi}(\mathbf{x}^{(i)}) \\
&= \begin{cases} 1 + \tilde{\boldsymbol{\theta}}^\top \tilde{\phi}(\mathbf{x}^{(i)}), & y^{(i)} = -1 \\ 1 - \tilde{\boldsymbol{\theta}}^\top \tilde{\phi}(\mathbf{x}^{(i)}), & y^{(i)} = +1 \end{cases} \\
&= 1 - y^{(i)} \, \tilde{\boldsymbol{\theta}}^\top \tilde{\phi}(\mathbf{x}^{(i)}).
\end{aligned}
$$

Roadmap

✓ Multiclass
one joint hinge loss

✓ Intercept (θ_0)
dummy feature 1

▶ Equivalence
g=2 $\Rightarrow$ binary (a)

Solution to 1 (b): Continued

From the per-term reduction, the sum collapses to a single term:

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \tfrac{1}{2}\|\boldsymbol{\theta}\|^2 + C\sum_{i=1}^{n} \max(1 - y^{(i)}\,\tilde{\boldsymbol{\theta}}^\top \tilde{\phi}(\mathbf{x}^{(i)}), 0)$$

$$= \tfrac{1}{2}\|\boldsymbol{\theta}\|^2 + C\sum_{i=1}^{n} \max(1 - y^{(i)}(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0).$$

multiclass ERM = binary ERM (a)

Roadmap

✓ Multiclass

one joint hinge loss

✓ Intercept (θ_0)

dummy feature 1

▶ Equivalence

g=2 ⇒ binary (a)

Solution to 1 (b): Continued

From the per-term reduction, the sum collapses to a single term:

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \tfrac{1}{2}\|\boldsymbol{\theta}\|^2 + C\sum_{i=1}^{n} \max(1 - y^{(i)}\,\tilde{\boldsymbol{\theta}}^\top \tilde{\phi}(\mathbf{x}^{(i)}), 0)$$

$$= \tfrac{1}{2}\|\boldsymbol{\theta}\|^2 + C\sum_{i=1}^{n} \max(1 - y^{(i)}(\boldsymbol{\theta}^\top \phi(\mathbf{x}^{(i)}) + \theta_0), 0).$$

multiclass ERM $=$ binary ERM (a) $\square$

Roadmap

- ✓ **Multiclass** — one joint hinge loss
- ✓ **Intercept (θ_0)** — dummy feature 1
- ▶ **Equivalence** — $g{=}2 \Rightarrow$ binary (a)

Exercise 1 (c)

Show that the regularized multiclass ERM problem in (b) can be written in the kernelized form:

$$\frac{1}{2}\boldsymbol{\beta}^\top \boldsymbol{K}\boldsymbol{\beta} + C\sum_{i=1}^{n}\sum_{y\neq y^{(i)}} \max(1 + (\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y} - (\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y^{(i)}}, 0),$$

where $\boldsymbol{\beta} \in \mathbb{R}^{ng}$ and $\boldsymbol{K} = \mathbf{X}\mathbf{X}^\top$ for $\mathbf{X} \in \mathbb{R}^{ng\times(d+1)}$ with row entries $\psi(\mathbf{x}^{(i)}, y)^\top$ for $i = 1, \ldots, n$, $y = 1, \ldots, g$, i.e.,

$$\mathbf{X} = \begin{pmatrix} \psi(\mathbf{x}^{(1)}, 1)^\top \\ \psi(\mathbf{x}^{(1)}, 2)^\top \\ \vdots \\ \psi(\mathbf{x}^{(1)}, g)^\top \\ \psi(\mathbf{x}^{(2)}, 1)^\top \\ \vdots \\ \psi(\mathbf{x}^{(n)}, g)^\top \end{pmatrix}.$$

Here, $(\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y}$ is the $((i-1)g+y)$-th entry of $\boldsymbol{K}\boldsymbol{\beta}$.

How to get there?

1. *Representer theorem:* $\boldsymbol{\theta}^* \in \mathrm{span}\{\psi(\mathbf{x}^{(i)}, y)\}$. Rewrite $\boldsymbol{\theta}$ with $\mathbf{X}$ and $\boldsymbol{\beta}$?
2. Then express $\frac{1}{2}\|\boldsymbol{\theta}\|^2$ via $\boldsymbol{K} = \mathbf{X}\mathbf{X}^\top$ and $\boldsymbol{\beta}$?
3. And the scores $\boldsymbol{\theta}^\top\psi(\mathbf{x}^{(i)}, y)$ via $\boldsymbol{K}\boldsymbol{\beta}$?

1 (c): Express $||\theta||^2$ with K and β

Representer theorem: $\boldsymbol{\theta}^*$ is a linear combination of the $\psi(\mathbf{x}^{(i)}, y)$, i.e.

$$\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\beta}, \qquad \boldsymbol{\beta} \in \mathbb{R}^{ng},$$

with the $\psi(\mathbf{x}^{(i)}, y)^\top$ as the rows of $\mathbf{X}$ (as on the previous slide). With $\boldsymbol{K} = \mathbf{X}\mathbf{X}^\top$:

$$||\theta||^2 = \theta^\top \theta = (\mathbf{X}^\top \beta)^\top \mathbf{X}^\top \beta = \boxed{\beta^\top \boldsymbol{K} \beta}.$$

Roadmap

✓ **Represent**
$\boldsymbol{\theta} = \mathbf{X}^\top \beta$

▶ **Norm**
$||\boldsymbol{\theta}||^2 = \boldsymbol{\beta}^\top \boldsymbol{K} \boldsymbol{\beta}$

Scores
$\boldsymbol{\theta}^\top \psi \to (\boldsymbol{K}\beta)$

1 (c): Express $||\theta||^2$ with K and β

Representer theorem: $\boldsymbol{\theta}^*$ is a linear combination of the $\psi(\mathbf{x}^{(i)}, y)$, i.e.

$$\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\beta}, \qquad \boldsymbol{\beta} \in \mathbb{R}^{ng},$$

with the $\psi(\mathbf{x}^{(i)}, y)^\top$ as the rows of $\mathbf{X}$ (as on the previous slide). With $\boldsymbol{K} = \mathbf{X}\mathbf{X}^\top$:

$$\|\boldsymbol{\theta}\|^2 = \boldsymbol{\theta}^\top \boldsymbol{\theta} = (\mathbf{X}^\top \boldsymbol{\beta})^\top \mathbf{X}^\top \boldsymbol{\beta} = \boxed{\boldsymbol{\beta}^\top \boldsymbol{K} \boldsymbol{\beta}}.$$

Roadmap

✓ **Represent**
$\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\beta}$

▶ **Norm**
$\|\boldsymbol{\theta}\|^2 = \boldsymbol{\beta}^\top \boldsymbol{K} \boldsymbol{\beta}$

Scores
$\boldsymbol{\theta}^\top \psi \to (\boldsymbol{K}\boldsymbol{\beta})$

1 (c): Express $\theta^\top \psi(\mathbf{x}^{(i)}, y)$ with K and β

Since $\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\beta}$: $\boldsymbol{\theta}^\top \psi(\mathbf{x}^{(i)}, y) = \boldsymbol{\beta}^\top \mathbf{X}\, \psi(\mathbf{x}^{(i)}, y)$ (scalar).

$$\begin{array}{r} \\ (i{-}1)g \\ \text{rows} \\ \\ \\ \\ \\ \end{array} \left[\begin{array}{c} \psi(\mathbf{x}^{(1)}, 1)^\top \\ \vdots \\ \psi(\mathbf{x}^{(i-1)}, g)^\top \\ \psi(\mathbf{x}^{(i)}, 1)^\top \\ \psi(\mathbf{x}^{(i)}, y)^\top \\ \psi(\mathbf{x}^{(i)}, g)^\top \end{array}\right] \begin{array}{l} \\ \\ \\ \\ \text{block } i \\ (g \text{ rows}) \\ \end{array}$$

So $\psi(\mathbf{x}^{(i)}, y)$ is row $(i{-}1)g{+}y$ of $\mathbf{X}$, i.e. *column* $(i{-}1)g{+}y$ of $\mathbf{X}^\top$.

Next: form $\mathbf{X}\psi(\mathbf{x}^{(i)}, y)$ via $\boldsymbol{K} = \mathbf{X}\mathbf{X}^\top$.

Roadmap

✓ **Represent**
$\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\beta}$

✓ **Norm**
$\|\boldsymbol{\theta}\|^2 = \boldsymbol{\beta}^\top \boldsymbol{K} \boldsymbol{\beta}$

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$\boldsymbol{\theta}^\top \psi \to (\boldsymbol{K}\boldsymbol{\beta})$

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Since $\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\beta}$: $\boldsymbol{\theta}^\top \psi(\mathbf{x}^{(i)}, y) = \boldsymbol{\beta}^\top \mathbf{X}\, \psi(\mathbf{x}^{(i)}, y)$ (scalar).

$$
\begin{array}{l}
\text{$(i-1)g$ rows}\left\{\begin{array}{c}\psi(\mathbf{x}^{(1)},1)^\top\\ \vdots\\ \psi(\mathbf{x}^{(i-1)},g)^\top\end{array}\right.\\
\qquad\qquad\quad\left.\begin{array}{c}\psi(\mathbf{x}^{(i)},1)^\top\\ \psi(\mathbf{x}^{(i)},y)^\top\\ \psi(\mathbf{x}^{(i)},g)^\top\end{array}\right\}\ \text{block } i\ (g\text{ rows})
\end{array}
$$

So $\psi(\mathbf{x}^{(i)}, y)$ is row $(i-1)g+y$ of $\mathbf{X}$, i.e. *column* $(i-1)g+y$ of $\mathbf{X}^\top$.

Next: form $\mathbf{X}\psi(\mathbf{x}^{(i)}, y)$ via $\boldsymbol{K} = \mathbf{X}\mathbf{X}^\top$.

Roadmap

✓ Represent
$\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\beta}$

✓ Norm
$\|\boldsymbol{\theta}\|^2 = \boldsymbol{\beta}^\top \boldsymbol{K} \boldsymbol{\beta}$

▶ Scores
$\boldsymbol{\theta}^\top \psi \to (\boldsymbol{K}\boldsymbol{\beta})$

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Since $\boldsymbol{\theta} = \mathbf{X}^\top\boldsymbol{\beta}$: $\quad \boldsymbol{\theta}^\top\psi(\mathbf{x}^{(i)}, y) = \boldsymbol{\beta}^\top\mathbf{X}\,\psi(\mathbf{x}^{(i)}, y)$ (scalar).

$$\begin{array}{rcl}
\begin{matrix}(i-1)g\\ \text{rows}\end{matrix}\left\{\begin{matrix}\\ \\ \\ \end{matrix}\right. & \left[\begin{matrix}\psi(\mathbf{x}^{(1)},1)^\top\\ \vdots\\ \psi(\mathbf{x}^{(i-1)},g)^\top\\ \psi(\mathbf{x}^{(i)},1)^\top\\ \psi(\mathbf{x}^{(i)},y)^\top\\ \psi(\mathbf{x}^{(i)},g)^\top\end{matrix}\right] & \begin{matrix}\\ \\ \\ \end{matrix}\left.\begin{matrix}\\ \\ \\ \end{matrix}\right\}\begin{matrix}\text{block } i\\ (g\text{ rows})\end{matrix}
\end{array}$$

So $\psi(\mathbf{x}^{(i)}, y)$ is row $(i-1)g+y$ of $\mathbf{X}$, i.e. *column* $(i-1)g+y$ of $\mathbf{X}^\top$.

Next: form $\mathbf{X}\psi(\mathbf{x}^{(i)}, y)$ via $\boldsymbol{K} = \mathbf{X}\mathbf{X}^\top$.

Roadmap

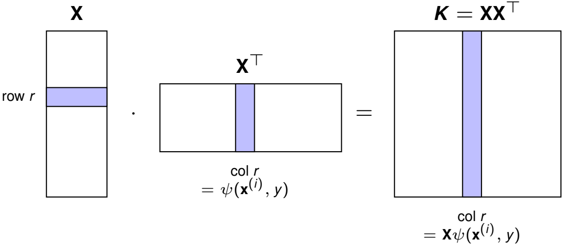
✓ **Represent**
$\boldsymbol{\theta} = \mathbf{X}^\top\boldsymbol{\beta}$

✓ **Norm**
$\|\boldsymbol{\theta}\|^2 = \boldsymbol{\beta}^\top\boldsymbol{K}\boldsymbol{\beta}$

▶ **Scores**
$\boldsymbol{\theta}^\top\psi \to (\boldsymbol{K}\boldsymbol{\beta})$

1 (c): $\mathbf{X}\,\psi(\mathbf{x}^{(i)}, y)$ is a column of $\boldsymbol{K}$

With $r := (i-1)g+y$: $\psi(\mathbf{x}^{(i)}, y) =$ col r of $\mathbf{X}^\top$, so $\mathbf{X}\psi(\mathbf{x}^{(i)}, y)$ is column r of $\mathbf{X}\mathbf{X}^\top$:



$\Rightarrow \mathbf{X}\psi(\mathbf{x}^{(i)}, y) =$ col r of $\boldsymbol{K}$ ($=$ row r, symmetric)

$\Rightarrow \ \boldsymbol{\beta}^\top \mathbf{X}\psi(\mathbf{x}^{(i)}, y) = (\boldsymbol{K}\boldsymbol{\beta})_r$.

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \tfrac{1}{2}\boldsymbol{\beta}^\top \boldsymbol{K}\boldsymbol{\beta} + \sum_{i=1}^{n} \sum_{y \neq y^{(i)}} \max\big(1 + (\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y} - (\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y^{(i)}}, 0\big).$$

Roadmap

✓ Represent

$\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\beta}$

✓ Norm

$\|\boldsymbol{\theta}\|^2 = \boldsymbol{\beta}^\top \boldsymbol{K}\boldsymbol{\beta}$

▶ Scores

$\boldsymbol{\theta}^\top \psi \to (\boldsymbol{K}\boldsymbol{\beta})$

1 (c): $\mathbf{X}\,\psi(\mathbf{x}^{(i)}, y)$ is a column of $\boldsymbol{K}$

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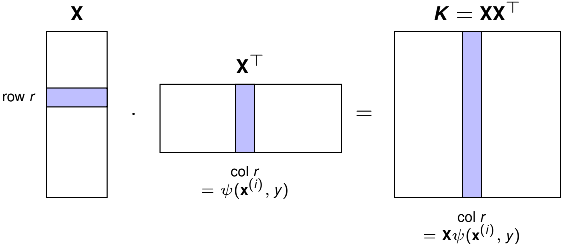

$\Rightarrow \mathbf{X}\psi(\mathbf{x}^{(i)}, y) =$ col r of $\boldsymbol{K}$ ($=$ row r, symmetric)
$\Rightarrow\ \boldsymbol{\beta}^\top \mathbf{X}\psi(\mathbf{x}^{(i)}, y) = (\boldsymbol{K}\boldsymbol{\beta})_r$.

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \tfrac{1}{2}\boldsymbol{\beta}^\top \boldsymbol{K}\boldsymbol{\beta} + \sum_{i=1}^{n} \sum_{y \neq y^{(i)}} \max\left(1 + (\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y} - (\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y^{(i)}},\ 0\right).$$

Roadmap

✓ **Represent**
$\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\beta}$

✓ **Norm**
$\|\boldsymbol{\theta}\|^2 = \boldsymbol{\beta}^\top \boldsymbol{K}\boldsymbol{\beta}$

▶ **Scores**
$\boldsymbol{\theta}^\top \psi \to (\boldsymbol{K}\boldsymbol{\beta})$

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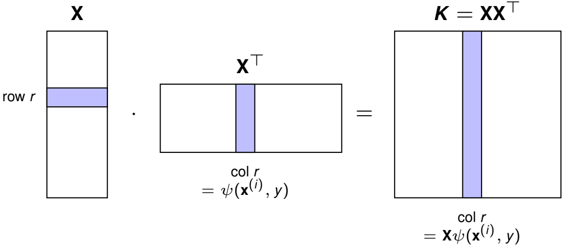
With $r := (i-1)g+y$: $\psi(\mathbf{x}^{(i)}, y) =$ col r of $\mathbf{X}^\top$, so $\mathbf{X}\psi(\mathbf{x}^{(i)}, y)$ is column r of $\mathbf{X}\mathbf{X}^\top$:



$\Rightarrow \mathbf{X}\psi(\mathbf{x}^{(i)}, y) =$ col r of $\boldsymbol{K}$ ($=$ row r, symmetric)

$\Rightarrow \boldsymbol{\beta}^\top\mathbf{X}\psi(\mathbf{x}^{(i)}, y) = (\boldsymbol{K}\boldsymbol{\beta})_r$.

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \tfrac{1}{2}\boldsymbol{\beta}^\top\boldsymbol{K}\boldsymbol{\beta} + \sum_{i=1}^{n}\sum_{y\neq y^{(i)}} \max\left(1 + (\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y} - (\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y^{(i)}},\, 0\right).$$

Roadmap

✓ Represent
$\boldsymbol{\theta} = \mathbf{X}^\top\boldsymbol{\beta}$

✓ Norm
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▶ Scores
$\boldsymbol{\theta}^\top\psi \to (\boldsymbol{K}\boldsymbol{\beta})$

Exercise 2: Kernel Trick

The polynomial kernel is defined as

$$k(x, \tilde{x}) = (x^\top \tilde{x} + b)^d.$$

Furthermore, assume that $x \in \mathbb{R}^2$ and $d = 2$. (a) Derive the explicit feature map ϕ taking into account that the following equation holds:

$$k(x, \tilde{x}) = \langle \phi(x), \phi(\tilde{x}) \rangle$$

Solution to 2 (a)

$$k(x, \tilde{x}) = (x^\top \tilde{x} + b)^2 = \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^\top \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} + b \right)^2$$

Solution to 2 (a)

$$
\begin{aligned}
k(x, \tilde{x}) &= (x^\top \tilde{x} + b)^2 = \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^\top \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} + b \right)^2 \\
&= (x_1 \tilde{x}_1 + x_2 \tilde{x}_2 + b)^2
\end{aligned}
$$

Solution to 2 (a)

$$
\begin{aligned}
k(x, \tilde{x}) &= (x^\top \tilde{x} + b)^2 = \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^\top \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} + b \right)^2 \\
&= (x_1\tilde{x}_1 + x_2\tilde{x}_2 + b)^2 \\
&= x_1^2\tilde{x}_1^2 + 2x_1\tilde{x}_1x_2\tilde{x}_2 + x_2^2\tilde{x}_2^2 + 2bx_1\tilde{x}_1 + 2bx_2\tilde{x}_2 + b^2
\end{aligned}
$$

Solution to 2 (a)

$$
\begin{aligned}
k(x, \tilde{x}) &= (x^\top \tilde{x} + b)^2 = \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^\top \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} + b \right)^2 \\
&= (x_1 \tilde{x}_1 + x_2 \tilde{x}_2 + b)^2 \\
&= x_1^2 \tilde{x}_1^2 + 2 x_1 \tilde{x}_1 x_2 \tilde{x}_2 + x_2^2 \tilde{x}_2^2 + 2 b x_1 \tilde{x}_1 + 2 b x_2 \tilde{x}_2 + b^2 \\
&= \left\langle \begin{pmatrix} x_1^2 \\ \sqrt{2} x_1 x_2 \\ x_2^2 \\ \sqrt{2b} x_1 \\ \sqrt{2b} x_2 \\ b \end{pmatrix}, \begin{pmatrix} \tilde{x}_1^2 \\ \sqrt{2} \tilde{x}_1 \tilde{x}_2 \\ \tilde{x}_2^2 \\ \sqrt{2b} \tilde{x}_1 \\ \sqrt{2b} \tilde{x}_2 \\ b \end{pmatrix} \right\rangle \\
&= \langle \phi(x), \phi(\tilde{x}) \rangle
\end{aligned}
$$

Exercise 2 (b)

(b) Describe the main differences between the kernel method and the explicit feature map.

Solution: Using the kernel method reduces the computational cost of the scalar product in the higher-dimensional feature space, since the feature map is never formed explicitly.

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(b) Describe the main differences between the kernel method and the explicit feature map.

Solution: Using the kernel method reduces the computational cost of the scalar product in the higher-dimensional feature space, since the feature map is never formed explicitly.

Summary — Connecting the exercise to the lecture

	What we derived	Lecture concept made concrete
1 (a)	Featurized soft-margin SVM = regularized ERM with the hinge loss: $\frac{1}{2}\|\boldsymbol{\theta}\|^2 + C\sum_{i=1}^{n}\max(1 - y^{(i)}(\boldsymbol{\theta}^\top\phi(\mathbf{x}^{(i)}) + \theta_0), 0)$.	Hinge loss; SVM as regularized empirical risk minimization.
1 (b)	Joint feature map $\psi(\mathbf{x}, y) = \frac{1}{2}\, y\, \tilde{\phi}(\mathbf{x})$, $\tilde{\phi} = (1, \phi)^\top$ (absorbs θ_0); the multiclass hinge collapses to the binary hinge when $g = 2$.	Multiclass SVM via a joint feature map $\psi(\mathbf{x}, y)$; multiclass hinge loss.
1 (c)	Representer theorem $\boldsymbol{\theta} = \mathbf{X}^\top\boldsymbol{\beta} \Rightarrow$ kernelized form $\frac{1}{2}\boldsymbol{\beta}^\top\boldsymbol{K}\boldsymbol{\beta} + \sum_{i=1}^{n}\sum_{y\neq y^{(i)}}\max(1 + (\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y} - (\boldsymbol{K}\boldsymbol{\beta})_{(i-1)g+y^{(i)}}, 0)$, $\boldsymbol{K} = \mathbf{X}\mathbf{X}^\top$.	Representer theorem; kernelization; Gram matrix $\boldsymbol{K}$.
2	Polynomial kernel $(x^\top\tilde{x} + b)^2 = \langle\phi(x), \phi(\tilde{x})\rangle$ with explicit $\phi \in \mathbb{R}^6$; the kernel returns the inner product without forming ϕ.	Kernel trick; kernels vs. explicit feature maps (compute savings).